



Emotion beliefs predict resilience via emotional reactivity: A multi-timepoint study in university students

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ABSTRACT

Understanding resilience in tertiary students is important, as this developmental period is characterised by elevated stressor exposure and vulnerability to mental health difficulties. Although emotion beliefs and emotional reactivity are each linked to mental health, little is known about their role in resilient functioning following adversity. This study used a three-wave longitudinal design in undergraduate university students ($N = 112$) to examine whether maladaptive emotion beliefs predict resilient functioning and whether these associations operate through positive and negative emotional reactivity. Emotion beliefs were assessed at baseline (T1), while emotional reactivity, stressor exposure, and mental health were assessed at two follow-up timepoints (T2 and T3). Resilient functioning was indexed relative to stressor exposure, with higher scores indicating better-than-expected emotional functioning. Results showed that positive emotional reactivity was positively associated with resilient functioning, whereas negative emotional reactivity was negatively associated with resilient functioning. Stronger beliefs that emotions are uncontrollable were associated with poorer resilient functioning overall, with some evidence of an indirect pathway through greater negative emotional reactivity. No corresponding associations were observed for beliefs about the usefulness of emotions. However, associations involving emotion beliefs were no longer significant after accounting for baseline mental health. The pattern of structural relationships did not differ significantly across follow-up assessments. These findings highlight emotional reactivity as an important correlate of resilient functioning and provide preliminary evidence that beliefs about emotion controllability may also be relevant, although their unique contribution beyond pre-existing mental health requires further investigation.

1. Introduction

Tertiary students face a range of persistent and acute stressors that threaten their psychological wellbeing (Pao, 2017). These include academic demands, financial insecurity, relational challenges, and uncertainties about identity and future pathways (Arnett, 2000). Such stressors are not only frequent but also cumulative, placing young adults at an elevated risk for adverse mental health outcomes. Indeed, emerging adulthood is a period marked by high prevalence of depression, anxiety, and stress-related disorders (Kessler et al., 2005; Thapar et al., 2022). Despite the near-ubiquity of stressor exposure, there is substantial variability in how individuals respond to these challenges. Some students manage to show better psychological functioning than

others who have experienced similar levels of adversity exposure, reflecting the concept of resilience. Understanding the mechanisms that account for these individual differences is essential to both theoretical accounts of resilience and the development of targeted support strategies within tertiary education settings (Marquez et al., 2023).

Understanding and supporting resilience in tertiary students is crucial because higher resilience is consistently associated with a range of adaptive outcomes, including lower rates of depression and anxiety, better academic performance, greater life satisfaction, and higher levels of psychological wellbeing (Cassidy, 2015; Hartley, 2011; Keye & Pidgeon, 2013). While improving student outcomes can and should involve efforts to reduce exposure to structural and interpersonal stressors such as financial hardship, discrimination, or academic

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pressure, such exposure cannot be entirely eliminated. Life inherently involves adversity, and students will inevitably face periods of challenge and uncertainty during and beyond their studies (Walters et al., 2025). Promoting resilience is therefore a vital parallel strategy to improving student outcomes. Moreover, resilient individuals are more likely to engage in prosocial, agentic behaviours that contribute to structural and cultural change (Liu et al., 2025). Thus, fostering student resilience is not only about improving individual mental health, it also strengthens the broader capacity of institutions and communities to face ongoing societal challenges (Esparza-Reig et al., 2022).

Psychological resilience, in this context, is defined as a better-than-expected mental health outcomes given the degree of adversity an individual has experienced (Kalisch et al., 2021). This conceptualisation reflects a departure from older trait-based definitions and aligns with outcome-focused models that emphasize the discrepancy between predicted and observed wellbeing (Chmitorz et al., 2018; Luthar et al., 2000; Southwick et al., 2014). Resilience can thus be operationalised through a residual approach, where mental health scores are regressed on adversity exposure and the residuals represent resilience (Cahill et al., 2022; Hölte & Ungar, 2022; Kalisch et al., 2021; Notebaert et al., 2024). A positive residual indicates that an individual is doing better than expected given their stressor load, whereas a negative residual indicates that an individual is showing poorer than expected mental health outcomes. This operationalisation captures the adaptive deviation from normative trajectories and is particularly suitable for longitudinal designs, where both stressor exposure and mental health are tracked over time (Kalisch et al., 2021).

The value of defining resilience in this way is both theoretical and practical. It allows for the identification of individual-level psychological variables that might predict why some students exhibit higher resilience than others. Kalisch et al. (2024) further propose a distinction between *resilience factors* and *resilience mechanisms*. Factors are more or less stable individual characteristics such as cognitive styles, beliefs, or personality traits that influence how stress is anticipated or appraised. Mechanisms, in contrast, are proximal processes that are engaged during or after stressor exposure and help buffer its impact. Resilience factors can be understood as predispositions that facilitate the activation of resilience mechanisms when the individual is under challenge. This framework can be applied to investigate candidate mechanisms and factors that may be implicated in individual differences in student resilience.

1.1. Emotion beliefs as a resilience factor

Within this framework, emotion beliefs represent a plausible candidate for a resilience factor. Emotion beliefs are relatively stable cognitive-affective schemas that shape how emotions are perceived, valued, and regulated (Becerra et al., 2020; Ford & Gross, 2019). Maladaptive beliefs, such as viewing emotions as harmful or unchangeable, are linked to reduced use of adaptive regulation strategies and greater psychological vulnerability (De Castella et al., 2013; Rimes & Chalder, 2010). These beliefs thus represent a framework that influences how emotional experiences are anticipated, interpreted, and responded to during stressor exposure, and consequently determine the emotional processes that are activated (Becerra et al., 2024; Ford & Gross, 2019). More adaptive emotion beliefs may facilitate the engagement of more effective resilience mechanisms, whereas maladaptive emotion beliefs may contribute to more intense, prolonged, or dysregulated emotional responses during adversity. In this way, the downstream effects of emotion beliefs can contribute to better or worse psychological outcomes in the face of stressor exposure.

Recent research has focused specifically on individuals' beliefs about the controllability and usefulness of emotions (Becerra et al., 2020). Adaptive emotion beliefs involve viewing emotions as useful and controllable, while maladaptive beliefs involve perceiving emotions as useless or uncontrollable. A growing body of research has linked

maladaptive emotion beliefs to poorer emotion regulation outcomes and higher levels of psychological distress (De Castella et al., 2013; Ford & Gross, 2019; Rimes & Chalder, 2010). Theoretically, these beliefs influence emotion regulation across multiple stages from the identification of emotions, to selecting regulation strategies, implementing these strategies, and evaluating their effectiveness (Gross, 2015). For example, individuals who view emotions as uncontrollable may be less likely to attempt regulation, use fewer or less effective strategies, and be less successful in modulating emotional experiences (Ford & Gross, 2019).

Despite the well-established association between emotion beliefs and mental health outcomes (Becerra et al., 2024), relatively little empirical research has examined the relationship between emotion beliefs and resilience in the context of adversity specifically. There is some indirect evidence that more maladaptive emotion beliefs may be associated with lower resilience. For example, Karnaze and Levine (2020) conducted an experiment in college students where one group of participants was encouraged to view emotions as helpful. Compared to a control group, these students showed greater emotional acceptance and faster recovery after a subsequent distressing film task. Other studies of young adults during real-world adversity echo these results. For example, a longitudinal study tracked first-year university students through the upheaval of the COVID-19 pandemic in 2020. Students with more adaptive emotion beliefs exhibited lower anxiety trajectories over the 6-month stress period than those with more maladaptive emotion beliefs (Waizman et al., 2023).

As such, given their demonstrated influence on emotional processes and indirect evidence of their contribution to resilience, we suggest that emotion beliefs may be conceptualised as a resilience factor, in that they may shape how individuals prepare for, perceive, and recover from adversity. Maladaptive emotion beliefs may predispose individuals to poor emotional responses in the face of stress, thereby increasing emotional strain and decreasing positive psychological outcomes.

1.2. Emotional reactivity as a resilience mechanism

With emotion beliefs as a candidate resilience factor, emotional reactivity becomes a candidate mechanism through which emotion beliefs may influence resilient outcomes. Defined as the threshold, intensity, and duration of emotional responses to stimuli, emotional reactivity is considered a proximal feature of emotional responding (Becerra et al., 2019; Davidson, 1998). Emotional reactivity is typically assessed using self-report measures such as the Perth Emotional Reactivity Scale (Becerra et al., 2019), and can be characterized separately for negative emotions (e.g. anger, fear) and positive emotions (e.g. joy, excitement).

Empirical research supports the role of emotional reactivity in mental health outcomes. For example, high negative reactivity has been associated with mood and anxiety disorders, while low positive reactivity has been linked to anhedonia and depressive symptoms (Bylsma et al., 2008; Nock et al., 2008). Emotional reactivity also varies across development, with young adults tending to exhibit greater reactivity than older adults, likely due to underdeveloped regulation strategies (Shapiro et al., 2016; Silvers et al., 2012).

Given the centrality of emotional responses to both momentary and chronic experiences of stress (Sun et al., 2023), emotional reactivity represents a key candidate resilience mechanism through which beliefs about emotions may influence longer-term psychological outcomes in students. Specifically, individuals who are highly emotionally reactive may experience more intense and persistent negative affect in response to daily and/or major stressors, thereby increasing their psychological toll. Conversely, high positive emotional reactivity may serve a protective function, promoting recovery from stress and fostering re-engagement with personal goals, social connection, and adaptive coping efforts (Tugade & Fredrickson, 2004).

1.3. The relationship between emotion beliefs, emotional reactivity, and resilience

Research with student populations suggests a clear link between maladaptive emotion beliefs and heightened negative emotional reactivity. For instance, a study of over 600 undergraduates used the Cognitive Mediation Beliefs Questionnaire (CMBQ) to assess beliefs in emotions as automatic “stimulus-response” (S-R) associations, versus the belief that emotions can be changed via cognition. The results showed that students with stronger S-R generation beliefs (i.e. thinking emotions are uncontrollable) reported significantly higher trait emotion reactivity. In contrast, those with stronger beliefs that emotions are controllable showed better emotional well-being and no elevated reactivity (Turner et al., 2024). This aligns with findings in the general population using the Emotion Beliefs Questionnaire (EBQ), which measures beliefs about the controllability and usefulness of positive and negative emotions. In this study, individuals who endorsed more maladaptive beliefs (agreeing that emotions are useless and uncontrollable) tended to have higher negative emotional reactivity and lower positive emotional reactivity. These correlations were especially pronounced for “uncontrollability” beliefs. Believing one cannot control negative emotions was a particularly strong correlate of heightened negative reactivity in this adult sample (Becerra et al., 2020).

Notably, the impact of these beliefs can extend to emotional outcomes in students. One laboratory study of undergraduates showed that maladaptive metacognitive beliefs (e.g. that thoughts and by extension emotions are uncontrollable) magnified the effect of emotional reactivity on anxiety levels (Clauss et al., 2020). In that study, students high in emotional reactivity became especially anxious after a negative mood induction if they also endorsed strong beliefs that they could not control their thoughts/emotions. This suggests that not only do maladaptive emotion beliefs relate to greater reactivity, they can also exacerbate the consequences of that reactivity on mental health.

As yet however, no studies have examined the combined role of emotion beliefs and emotional reactivity in individual differences in student resilience. The current study addresses this gap using longitudinal data collected across a university semester. Specifically, we examined whether emotion beliefs assessed at the beginning of the semester predict resilience outcomes later in the semester, and whether these effects are mediated by emotional reactivity. By modelling these relationships across two follow-up assessments, we were also able to examine whether the pattern of structural associations differed across timepoints.

1.4. The current study

The aim of the current study was to test the novel hypothesis that maladaptive emotion beliefs predict poorer resilience in tertiary students, and that this relationship can be explained in part by emotional reactivity. Specifically, we hypothesised that students who believe emotions are uncontrollable and useless will demonstrate poorer resilience, and this effect will be mediated by their increased negative and reduced positive emotional reactivity.

The study used a three-wave longitudinal design with data collected through the Perth Resilience Project, which assessed undergraduate university students on a range of variables across multiple timepoints in a university semester. Emotion beliefs were assessed at baseline (T1), consistent with their conceptualisation as relatively stable individual difference factors that shape downstream emotional responding (Ford & Gross, 2019). Current emotional reactivity, stressor exposure, and mental health were assessed at two follow-up time points (T2 and T3). Emotional reactivity was conceptualised as a proximal resilience mechanism occurring during or shortly following stressor exposure (Kalisch et al., 2014), and was therefore examined in relation to resilient functioning within the same assessment period. A resilient functioning index (RFI) was computed in line with the Frequent Stressor and Health

monitoring -FRESHMO- paradigm (Kalisch et al., 2021), by regressing mental health scores on measures of stressor exposure at each assessment point.

The novel hypothesis under test predicts that i. more maladaptive emotion beliefs at baseline will be associated with lower resilience at the later timepoints; ii. current emotional reactivity will mediate the relationship between emotion beliefs and resilience, such that maladaptive emotion beliefs at baseline will predict higher negative reactivity and lower positive reactivity at the later timepoints, which in turn will predict lower resilience.

To assess the similarity in the pattern of associations across follow-up assessments, we modelled the hypothesised relationships at T2 and T3 simultaneously within a single structural equation model. This approach accounted for the repeated-measures structure of the data by estimating cross-wave residual covariances between corresponding measures of emotional reactivity and resilient functioning at T2 and T3. We compared an unconstrained model, in which corresponding structural paths were freely estimated at each follow-up assessment, with a constrained model in which these paths were set equal across T2 and T3. This comparison allowed us to examine whether the strength of the hypothesised associations differed across follow-up assessments while accounting for within-person dependencies between repeated measures.

By integrating emotion beliefs and emotional reactivity into a unified resilience model, this study contributes to a more in-depth understanding of the psychological processes that support resilient functioning in young adults experiencing the stressors associated with university life.

2. Methods

2.1. Participants

Participants were recruited for the study through the School of Psychological Science Research Participation System at the University of Western Australia, which advertises Psychology studies to students enrolled in specific psychology undergraduate units. All participants received course credit. After data cleaning, the final sample used in the hypothesis test consisted of 112 participants. Ages ranged from 17 to 25 years ($M = 18.76$, $SD = 1.56$). The gender identity of the sample included 91 female participants, 20 male, and one non-binary. Most participants reported their ethnic backgrounds as Asian (36.6%), mixed (25.0%), and Australian (19.6%). Additionally, most participants were first year (73.0%) and domestic (80.0%) students. The average socioeconomic status ranking (see *Demographics* below) was 5.67, with a range of 1 to 8, and most (75.89%) ranking between 5 and 7.

2.2. Measures

2.2.1. Demographics

Demographic data was recorded for descriptive purposes. The demographics included questions on student age, ethnicity, gender identity, domestic or international student status, and current year of study. Subjective social status was measured using the MacArthur Scale of Subjective Social Status (Adler et al., 2000). Participants were also asked to rank where they stand in society, by placing themselves on a scale numbered from 1 (who have the lowest standing in their society) to 10 (who have the highest standing in their society).

2.2.2. Emotion beliefs

The Emotions Belief Questionnaire (EBQ) (Becerra et al., 2020) was used to assess individuals' beliefs about the usefulness and controllability of positive and negative emotions. The 16-item measure includes four subscales: negative-controllability (e.g. “People cannot learn techniques to effectively control their negative emotions”), positive-controllability (e.g. “People cannot control their positive emotions”), negative-usefulness (e.g. “Negative emotions are harmful”), and

positive-usefulness (e.g. “Positive emotions are very unhelpful to people”), with four items per subscale. Positive and negative subscales can be combined to form overall uselessness and uncontrollability subscales (Becerra et al., 2020).

Respondents are asked to indicate how much they agree with each statement on a 7-point Likert scale ranging from one (*strongly disagree*) to seven (*strongly agree*). Uncontrollability and uselessness subscale scores are calculated by summing item scores within subscales. Higher scores reflect stronger beliefs that emotions (negative and positive) are uncontrollable and useless. The uncontrollability and uselessness subscales showed good reliability (Cronbach's $\alpha = .84$ and 0.81 , respectively), and previous research has supported its validity, with strong evidence of cross-cultural validity (Johnston et al., 2024; Ranjbar et al., 2023).

2.2.3. Emotional reactivity

The Perth Emotional Reactivity Scale (PERS) (Becerra et al., 2019) was used to assess emotional reactivity states. The scale measures the activation (e.g. “I tend to get happy very easily”), intensity (e.g. “If I was upset, I felt it more intensely than everyone else”) and duration (e.g. “When I was happy, the feeling stayed with me for a while”) of positive and negative emotions within a given timeframe. It includes six subscales: positive-activation (PA), positive-intensity (PI), positive-duration (PD), negative-activation (NA), negative-intensity (NI), and negative-duration (ND), each with five items. Participants were asked to rate how much the statements applied to them during the past two weeks. Items are scored on a 5-point Likert scale ranging from one (*not at all*) to five (*almost always*). Items within each subscale were summed to create positive (PA + PD + PI) and negative (NA + ND + NI) reactivity subscale scores, with higher scores indicating higher emotional reactivity. The subscales scores demonstrated excellent reliability at each timepoint (Cronbach's α ranging from 0.94 to 0.96) and the scale has shown good validity (Preece et al., 2019).

2.2.4. Resilience

To measure resilience as an outcome, we calculated a Resilience Functioning Index (RFI) based on the Frequent Stressor and Mental Health Monitoring (FRESHMO) paradigm (Kalisch et al., 2021). To derive this RFI, we incorporated measures of stressor exposure and mental health.

2.2.4.1. Stressor exposure. Stressor exposure was assessed using three instruments which capture exposure to various potential stressors, ranging from daily hassles to major traumatic events as recommended by Kalisch et al. (2021).

The Mainz Inventory of Microstressors (MIMIS) (Chmitorz et al., 2020) consists of 58 microstressors (i.e., annoyances and hassles) that may occur in daily life. This study used an adapted and shortened version, including the ten most frequent and the ten most severe microstressors (as reported by Chmitorz et al., 2020). Two of these items overlapped, resulting in final 18 items (items 11, 38, 6, 5, 50, 55, 35, 33, 51, 42, 23, 15, 49, 32, 56, 19, 58 of the full-length survey). An example item is, “Interruption during an activity (e.g., at work or during leisure activities)”. Participants reported the frequency of each microstressor over the past seven days. Items are scored on an 8-point Likert scale with responses corresponding to the number of days occurred, ranging from zero (*did not occur*) to seven (*seven days*). Total scores are calculated by summing item scores, with higher scores indicating more microstressor exposure. The MIMIS has shown good validity (Chmitorz et al., 2020).

The Trauma History Questionnaire (THQ) (Hooper et al., 2011) assesses individuals' trauma history through 18 questions related to crime, general disaster, and physical and sexual experiences (e.g. “has anyone attempted to rob you or actually robbed you (i.e., stolen your personal belongings)”). Participants were asked to indicate whether they had experienced each traumatic event since the last assessment point (approximately a three week interval). The number of yes answers

represents the total score, with higher scores indicating more trauma exposure. The THQ is a valid measure of trauma history (Hooper et al., 2011).

The Negative Life Event Scale for Students (NLESS) (Buri et al., 2018) is a 25-item negative events checklist specifically relevant to the student population (e.g. experiencing financial strain). Participants were asked to indicate whether they had experienced each negative life event since the last assessment point. The number of yes answers represents the total score, with higher scores indicating more negative life event exposure. The NLESS has been shown a valid measure of negative life events for students (Buri et al., 2018).

2.2.4.2. Mental health. To index mental health, researchers have suggested using a dimensional and transdiagnostic approach, capturing the absence of emotional difficulties in the domains of anxiety, depression, and stress (Kalisch et al., 2014, 2020). However, it is also known that positive and negative emotions independently contribute to people's overall emotional experience (Watson & Naragon, 2009), and according to the broaden-and-build theory, positive emotions may promote coping with adversity and thus positively contribute to resilient functioning (Fredrickson, 2001). As such, in the current study, mental health was assessed transdiagnostically by including measures of depression, anxiety, perceived stress, and experiences of positive emotions. Each of the four total scores from the measures below were first standardised and then summed (with the positive emotions measure reverse-scored) to produce a mental health score, with higher scores indicating poorer mental health.

The 9-item Patient Health Questionnaire (PHQ-9) (Kroenke et al., 2001) is a widely used measure of depression. Respondents report the frequency of symptoms over the past two weeks (e.g. “Little interest or pleasure in doing things”) on a 4-point Likert scale ranging from zero (*not at all*) to three (*nearly every day*). Total scores are calculated by summing item scores, with higher scores indicating more depressive symptoms. The PHQ-9 demonstrated excellent reliability (Cronbach's $\alpha = .89$ at each timepoint) and has shown good validity in young people (Fonseca-Pedrero et al., 2023).

The 7-item Generalised Anxiety Disorder scale (GAD-7) (Spitzer et al., 2006) asks respondents to report on their anxiety symptoms over the past two weeks (e.g. “Being so restless that it's hard to sit still”), on a 4-point Likert scale ranging from zero (*not at all*) to three (*nearly every day*). Total scores are calculated by summing item scores, with higher scores indicating more anxiety symptoms. The GAD-7 showed strong reliability at each timepoint (Cronbach's $\alpha = .91$ and 0.92) and is a valid measure of anxiety symptoms (Löwe et al., 2008).

The 10-item Perceived Stress Scale (PSS-10) (Cohen et al., 1983) asks respondents to indicate how often they felt or thought a certain way over the past month (e.g. “... felt that you were unable to control the important things in your life?”). Items are scored on a 5-point Likert scale ranging from zero (*never*) to four (*very often*). After reverse scoring negatively worded items, item scores are summed, with higher scores indicating more frequently perceived stress. The PSS-10 demonstrated good reliability at each timepoint (Cronbach's $\alpha = 0.82$ and 0.87) and has shown good validity (Roberti et al., 2006).

The Dispositional Positive Emotion Scales (DPES) (Shiota et al., 2006) is a 38-item self-report questionnaire measuring respondents' identification with positive emotions, including seven subscales for joy, contentment, pride, love, compassion, amusement, and awe (e.g. “I felt awe”). The current study used an adapted version shortened to three items per subscale (items 1, 3, 4, 7, 10, 11, 12, 13, 14, 19, 20, 21, 23, 26, 27, 28, 29, 30, 33, 34, 35 of the full-length survey). To measure the experience of positive emotion as a transient state, the item scale was changed to ask respondents to rate how often they experienced each item over the past two weeks, on a scale of 0 (*not at all*) to 3 (*nearly every day*). Item scores are summed to create a total score, with higher scores indicating more frequent experiences of positive emotions. The scale has

good validity (Dixon et al., 2018), and our shortened version showed excellent internal consistency at each timepoint (Cronbach's $\alpha = 0.94$ and 0.96).

2.2.4.3. Calculating the resilient functioning index. The FRESHMO paradigm assumes a positive linear relationship between stressor exposure (S) and experiences of mental health difficulties (M) over the course of an observation period, and a participant's deviation from the sample's normative S-M relationship (the regression residual) can be used to represent a resilience score (Kalisch et al., 2021). As such, stressor exposure was linearly regressed against total mental health difficulties scores. Stressor exposure was modelled using two predictors reflecting conceptually distinct sources of adversity: exposure to major life stressors (operationalised as the summed THQ and NLESS scores) and exposure to microstressors (MIMIS). THQ and NLESS were summed to preserve their shared count-based interpretation of cumulative major stressor burden, while MIMIS was retained as a separate predictor because it captures a qualitatively different form of stress exposure. Including these predictors separately allowed the regression model to estimate their relative contributions to mental health difficulties rather than imposing equal weighting via standardisation or averaging. The resulting residuals therefore index resilient functioning relative to expected mental health outcomes given both major and micro level stressor exposure. Residuals were reverse scored, such that a positive resilient functioning index (RFI) indicates more resilient outcomes as the participant is experiencing fewer mental health difficulties than what is predicted based on stressor exposure. A negative RFI indicates less resilient outcomes as compared to the sample.

2.3. Procedure

This study was part of a bigger project named the “Perth Resilience Project”, which was conducted in the first half of 2024 (see also Notebaert et al., 2026). Participants accessed the study online via a Qualtrics link. They were asked to complete it in a quiet environment where they would not be interrupted. The study consisted of three assessment points. Baseline data (T1) was recorded in the first three weeks of a 12-week semester. This baseline assessment included the demographics questionnaire and the assessment of emotion beliefs (EBQ) among a range of other measures including mental health surveys (totalling 1 h in duration). The second assessment point (T2) opened two weeks later with two-week deadline for participation, and included the emotional reactivity measure (PERS) as well as the stressor exposure and mental health measures. The third assessment point (T3) again opened two weeks later with a two-week participation deadline, and included the same emotional reactivity, stressor exposure, and mental health measures. These latter two sessions took approximately 45 min. Participants were sent email notifications at the start of T2 and T3, as well as two reminder emails during each active assessment period. Informed consent was provided at each assessment point, and participants were debriefed at the end of the last session. This project was approved by the Human Research Ethics Office at the University of Western Australia. The dataset can be found at the following link: https://osf.io/6n8g5/overview?view_only=535ffc0174b3490cb5fbc6278a58e9e5.

3. Results

3.1. Data processing

Of 848 eligible students, the number of students who participated in this study was 221 at T1, 182 at T2, and 158 at T3. In the data cleaning process, first data from 20 participants who were older than 25 were removed to reduce developmental heterogeneity and retain a sample representative of traditional-aged undergraduate students (Chung et al., 2017). Then, data from participants who did not complete any

questionnaires were deleted ($n = 1$ at T1, $n = 6$ at T2, $n = 3$ at T3). Attrition analyses compared participants who completed all study waves with those who discontinued participation on key baseline demographic and study variables, including age, gender, domestic/international status, first-year status, baseline mental health, stressor exposure, resilience, and emotion beliefs. No significant differences emerged between completers and non-completers on any variable (all $ps > 0.14$), suggesting that attrition was not systematically related to the primary study constructs.

To screen for careless responding, a measure of response time at the survey page level was used (Meade & Craig, 2012; Ward & Meade, 2018). A cut-off of 2 s per item has been recommended for standard measures (Meade & Craig, 2012). However, for repeated measures Jaso et al. (2022) have suggested a more liberal approach, and allowing for less conservative cutoffs as completion efficiency increases. In addition, as checklist type items take less time to complete than items with Likert response scales, different cutoffs were used for these two types of measures. As a result, for T1 data, data from participants who on average responded faster than 1.5 s per item for Likert scales responses, or faster than 1 s per items for checklist responses were removed. For both T2 and T3 these cutoffs were 1.25 s for Likert scale responses and 0.75 s for checklist responses. This resulted in the removal of 10 additional participants at T1, 10 at T2, and 18 at T3. Of the 190 remaining participants at T1, 41 did not participate in both subsequent sessions and a further 37 did not participate in the T3 session. As a result, after joining data of participants who had completed all three phases, the final sample consisted of 112 participants.

Because the SEM models used composite scale scores as manifest variables, item-level Multiple Imputation by Chained Equations (MICE) was used as a scale-score procedure to derive composite scores for partially completed questionnaires, whereas FIML was used to handle missingness in the resulting scale-level variables. As such, prior to analysis, partial item-level missing data were examined and imputed using MICE with predictive mean matching (PMM). Imputation was conducted only when participants provided partial responses on a measure. Cases with all items missing for a measure were not imputed and remained coded as missing. Due to the nature of the adversity checklists, no missing data were imputed for these measures. The imputation procedure used five iterations ($\text{maxit} = 5$) and assumed data were missing at random (MAR) (Schafer & Graham, 2002). Remaining missingness at the variable level, including participants with entirely missing measures, was subsequently handled in the SEM analyses using full information maximum likelihood (FIML) estimation in *lavaan* (Rosseel, 2012). Standard errors and 95% confidence intervals for model parameters were estimated using nonparametric bootstrap resampling with 10,000 bootstrap draws.

Descriptive statistics for the adversity and mental health surveys that were used to create the composite measures of baseline mental health at T1, and the Resilient Functioning Index at T2 and T3 can be found in Table 1. Baseline PHQ-9 scores indicated elevated depressive symptomatology in the current sample, with average scores falling within the mild-to-moderate severity range. Of the sample, 22% scored within the minimal range (0–4), 33% within the mild range (5–9), 20% within the moderate range (10–14), 17% within the moderately severe range (15–19), and 8% within the severe range (20+). Relative to normative data from similarly aged community samples, the present sample showed substantially elevated depressive symptoms. In population-based norms for individuals aged 16–24 years, approximately 75% score within the minimal range, approximately 16% within the mild range, and only around 5% score within the moderate-to-severe range (scores > 10 ; Kliem et al., 2024). Similarly, the average GAD7 score was in the mild anxiety range and 34% of the sample scored 10 or higher, which is considered a reasonable cut-off when screening for anxiety disorders. Compared to college student population norms, these scores indicate heightened levels of anxiety (Byrd-Bredbenner et al., 2021).

Table 1

Descriptive statistics for adversity exposure and mental health measures used to derive baseline mental health and the Resilient Functioning Index (RFI) at T2 and T3.

	Mean	SD	Min	Max
<i>Baseline Mental Health</i>				
T1 PHQ9	9.79	6.15	0	25
T1 GAD7	8.18	5.42	0	21
T1 PSS10	20.48	5.42	6	35
T1 S-DPES	30.16	11.40	7	53
<i>T2 Resilient Functioning Index</i>				
T2 MIMIS	32.15	15.69	0	77
T2 NLESS since T1	1.71	3.53	0	25
T2 THQ since T1	0.26	0.82	0	7
T2 PHQ9	9.29	5.99	0	27
T2 GAD7	7.83	5.29	0	21
T2 PSS10	19.42	7.13	3	39
T2 S-DPES	29.23	11.84	6	61
<i>T3 Resilient functioning Index</i>				
T3 MIMIS	34.33	14.41	0	90
T3 NLESS since T2	1.97	4.25	0	25
T3 THQ since T2	0.58	3.20	0	24
T3 PHQ9	9.99	6.31	0	27
T3 GAD7	7.98	5.46	0	21
T3 PSS10	20.14	6.93	5	36
T3 S-DPES	28.74	14.29	2	61

Note. PHQ9 = Patient Health Questionnaire; GAD7 = Generalised Anxiety Disorder Scale; PSS10 = Perceived Stress Scale; S-DPES = State version of the Dispositional Positive Emotion Scales; MIMIS = Mainz Inventory of Microstressors; NLESS = Negative Life Events Scale for Students; THQ = Trauma History Questionnaire.

3.2. Descriptives

The descriptive statistics of each of the key variables in the statistical analyses are described in Table 2. All variables satisfied normality assumptions (Skewness $<|2|$ Kurtosis $<|7|$; Field, 2013). To identify multivariate outliers in the dataset, Mahalanobis distances were computed for each observation. The critical threshold for outlier detection was determined using a chi-squared distribution, with a 99.9% confidence level. None of the observations had Mahalanobis distances exceeding this threshold; therefore, no multivariate outliers were identified.

3.3. Relationships between key variables

The zero-order correlations between the key variables are presented in Table 3. Notably, correlations between beliefs of uselessness and uncontrollability and emotional reactivity are absent or small in magnitude, with the strongest relationships observed at T3. Both positive and negative emotional reactivity exhibit strong associations with the RFI at both time points. Additionally, the correlations between time points of the emotional reactivity and resilience indices are large, indicating a degree of stability in these variables over time.

Table 2

Descriptive statistics for key variables.

Variable	n	Mean	SD	Min	Max	Skew	Kurtosis
T1 Uselessness Beliefs	112	16.98	5.89	8	34	0.38	-0.21
T1 Uncontrollability Beliefs	112	20.04	6.89	8	38	0.29	-0.41
T2 Negative Reactivity	112	39.28	13.67	16	74	0.33	-0.77
T2 Positive Reactivity	112	42.3	12.25	16	70	0.10	-0.76
T3 Negative Reactivity	112	39.22	14.09	15	70	0.50	-0.66
T3 Positive Reactivity	112	41.75	13.96	15	71	0.10	-1.03
T2 RFI	112	-0.06	0.95	-2.31	1.91	-0.12	-0.51
T3 RFI	109	-0.06	1.01	-2.06	2.53	0.15	-0.51

T1 = Time 1, T2 = Time 2, T3 = Time 3. Uncontrollability and Uselessness beliefs scores were derived from the EBQ. Positive and negative reactivity scores were derived from the PERS. RFI = Resilient Functioning Index.

3.4. Testing the proposed hypothesis

To test the predictions generated by the hypothesis, we fit a single-group structural equation model (SEM) in which the hypothesised relationships at T2 and T3 were modelled simultaneously (see Fig. 1). Baseline emotion beliefs (T1 EBQ Uncontrollability and EBQ Uselessness) were specified as exogenous predictors. At each follow-up assessment, direct paths were estimated from emotion beliefs to positive and negative emotional reactivity (PERS) and to the Resilient Functioning Index (RFI), as well as from positive and negative emotional reactivity to the RFI. Cross-wave residual covariances were estimated between corresponding T2 and T3 measures of positive reactivity, negative reactivity, and the RFI to account for within-person dependencies across repeated assessments. Indirect effects from each emotion belief to the RFI via positive and negative emotional reactivity were estimated, and total effects were calculated as the sum of the corresponding direct and indirect effects.

We first estimated an unconstrained model in which corresponding structural paths were freely estimated at T2 and T3, followed by a constrained model in which corresponding paths were set equal across the two follow-up assessments. The cross-wave residual covariances were freely estimated in both models. The models were compared using a chi-square difference test to evaluate whether constraining the structural relationships to equality significantly worsened model fit. AIC and BIC were also compared to assess relative parsimony. Model fit was evaluated using χ^2 , CFI, RMSEA (95% CI), and SRMR.

The chi-square difference test comparing the unconstrained and constrained models revealed no statistically significant difference in model fit, $\chi^2(8) = 12.929$, $p = .114$, suggesting no strong evidence that the structural relationships differed across the two follow-up assessments. The constrained model also showed lower AIC (3857.1 vs. 3860.2) and BIC (3919.7 vs. 3944.5) values compared to the unconstrained model, favouring the more parsimonious constrained solution. Given that constraining paths to equality did not significantly worsen model fit, the constrained model was used as the primary basis for interpretation.

The constrained model demonstrated excellent fit, $\chi^2(16) = 21.003$, $p = .178$, and a comparative fit index (CFI) of 0.987, indicating excellent model fit. The root mean square error of approximation (RMSEA) was 0.053 (90% CI 0.000 - 0.108), indicating acceptable point-estimate fit. The standardized root mean square residual (SRMR) was 0.055, which is well below the recommended threshold of 0.08. The path coefficients can be observed in Table 4.

Across both follow-up assessments, positive emotional reactivity was positively associated with resilient functioning ($B = 0.029$, $p < .001$), whereas negative emotional reactivity was negatively associated with resilient functioning ($B = -0.038$, $p < .001$). Neither uncontrollability nor uselessness beliefs showed statistically significant direct associations with resilient functioning or with positive or negative emotional reactivity based on the Wald tests. However, the bootstrap confidence interval for the association between uncontrollability beliefs and negative reactivity excluded zero ($B = 0.403$, 95% CI [0.014, 0.876]), providing

Table 3
Correlations between measures of emotion beliefs, emotional reactivity, and resilience.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
(1) T1 Uselessness Beliefs	1						
(2) T1 Uncontrollability Beliefs	0.41	1					
(3) T2 Negative Reactivity	-0.06	0.10	1				
(4) T2 Positive Reactivity	-0.11	-0.15	-0.09	1			
(5) T3 Negative Reactivity	-0.03	0.21*	0.75**	-0.17	1		
(6) T3 Positive Reactivity	-0.10	-0.22*	-0.06	0.78**	-0.09	1	
(7) T2 RFI	-0.11	-0.17	-0.59**	0.49**	-0.51**	0.41**	1
(8) T3 RFI	-0.21*	-0.41**	-0.43**	0.44**	-0.60**	0.46**	0.67**

Note * $p < .05$, ** $p < .001$. T1 = Time 1, T2 = Time 2, T3 = Time 3. Uncontrollability and Uselessness beliefs scores were derived from the EBQ. Positive and negative reactivity scores were derived from the PERS. RFI = Resilient Functioning Index.

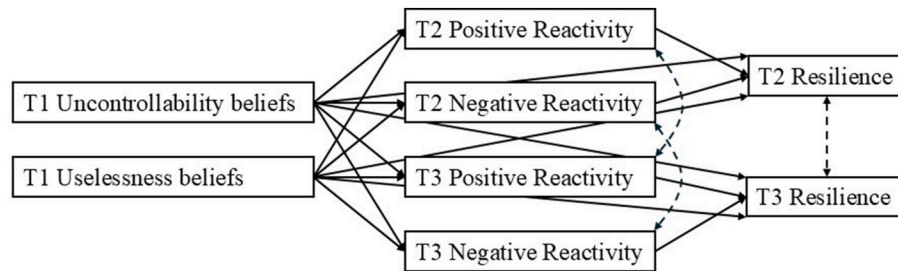


Fig. 1. Structural model examining associations between baseline emotion beliefs, emotional reactivity, and resilient functioning across T2 and T3.

Note. Single-headed arrows represent hypothesised structural paths. Double-headed dashed arrows represent freely estimated cross-wave residual covariances between corresponding measures at T2 and T3. T1 emotion beliefs were specified as exogenous predictors.

some evidence that stronger beliefs that emotions are uncontrollable were associated with greater negative emotional reactivity. Similarly, the bootstrap confidence interval for the indirect effect of uncontrollability beliefs on resilient functioning through negative reactivity excluded zero ($B = -0.015$, 95% CI $[-0.033, -0.001]$), providing evidence for this hypothesised indirect pathway. None of the other hypothesised indirect effects had confidence intervals that excluded zero. The total effect of uncontrollability beliefs on resilient functioning was also significant ($B = -0.037$, 95% CI $[-0.068, -0.007]$, $p = .018$), indicating that stronger beliefs that emotions are uncontrollable were associated with poorer resilient functioning overall.

Additional sensitivity analyses were conducted controlling for baseline mental health (see Appendix A). In these models, baseline mental health significantly predicted subsequent emotional reactivity and resilient functioning. After accounting for baseline mental health, neither uncontrollability nor uselessness beliefs significantly predicted emotional reactivity or resilient functioning, and none of the hypothesised indirect effects were statistically significant. These findings suggest that emotion beliefs did not explain unique variance in subsequent emotional reactivity or resilient functioning beyond that accounted for by baseline mental health.

4. Discussion

The aim of the current study was to examine whether maladaptive emotion beliefs, specifically beliefs that emotions are useless and uncontrollable, may represent resilience factors associated with poorer resilient functioning in tertiary students, and whether these relationships may be explained in part by emotional reactivity as a resilience mechanism. Using a three-wave longitudinal design, we modelled these relationships simultaneously across two follow-up assessments while accounting for within-person dependencies across repeated measures. The results provided limited support for our hypotheses. Positive and negative emotional reactivity were strongly associated with resilient functioning in the expected directions. Stronger baseline beliefs that emotions are uncontrollable also showed a significant overall association with poorer resilient functioning. The bootstrap confidence

intervals of the individual paths from uncontrollability beliefs to negative emotional reactivity and the corresponding indirect effect just excluded zero, providing some evidence consistent with the hypothesised pathway whereby uncontrollability beliefs are associated with greater negative emotional reactivity and, in turn, poorer resilient functioning. No corresponding evidence emerged for positive emotional reactivity or for beliefs regarding the usefulness of emotions. The pattern of structural relationships did not differ significantly across the two follow-up assessments. However, sensitivity analyses indicated that the associations involving emotion beliefs were substantially attenuated after accounting for baseline mental health, suggesting that emotion beliefs did not explain unique variance in subsequent emotional reactivity or resilient functioning beyond baseline mental health.

An important finding was the differing pattern of associations observed for uncontrollability and usefulness beliefs. The primary analyses provided some evidence that beliefs that emotions are uncontrollable may be associated with poorer resilient functioning through greater negative emotional reactivity, though the magnitude of these associations was small. However, these associations were substantially attenuated after controlling for baseline mental health, suggesting that their relationship with subsequent emotional reactivity and resilient functioning may be partly accounted for by broader emotional distress or vulnerability already present at baseline. This interpretation aligns with previous research demonstrating substantial overlap between maladaptive emotion beliefs and psychopathology symptoms (Kneeland et al., 2016). In contrast, beliefs regarding the usefulness of emotions showed neither direct nor indirect associations with resilient functioning. Together, these findings suggest that some dimensions of emotion beliefs may be more relevant to resilient functioning than others, and future research may benefit from examining controllability and usefulness beliefs separately.

The observed association between controllability beliefs and resilient functioning is consistent with previous research demonstrating associations between maladaptive beliefs about emotion and poorer mental health (Becerra et al., 2024; Kneeland et al., 2016), as well as evidence suggesting that more adaptive beliefs about emotions are associated with greater stress recovery (Karnaze & Levine, 2020; Waizman et al.,

Table 4

Unstandardized path coefficients, bootstrap standard errors (SE), 95% confidence intervals (CI), z-values, and p-values for the constrained structural equation model examining associations between baseline (T1) emotion beliefs, follow-up emotional reactivity, and follow-up resilient functioning across T2 and T3 assessments.

Path	B	SE	95% CI	z	p
<i>Direct Effects on Resilient Functioning Index</i>					
Uncontrollability Beliefs	−0.013	0.009	[−0.032, 0.005]	−1.334	0.182
Uselessness Beliefs	−0.019	0.011	[−0.040, 0.003]	−1.691	0.091
Positive Reactivity	0.029	0.004	[0.021, 0.037]	6.937	< 0.001
Negative Reactivity	−0.038	0.004	[−0.046, −0.030]	−9.693	< 0.001
<i>Paths Predicting Emotional Reactivity</i>					
Uncontrollability Beliefs → Positive Reactivity	−0.301	0.183	[−0.643, 0.072]	−1.649	0.099
Uncontrollability Beliefs → Negative Reactivity	0.403	0.218	[0.014, 0.876]	1.846	0.065
Uselessness Beliefs → Positive Reactivity	−0.054	0.213	[−0.482, 0.352]	−0.254	0.799
Uselessness Beliefs → Negative Reactivity	−0.281	0.236	[−0.753, 0.176]	−1.189	0.235
<i>Indirect Effects</i>					
Uncontrollability Beliefs → Pos. Reactivity → RFI	−0.009	0.006	[−0.020, 0.002]	−1.564	0.118
Uncontrollability Beliefs → Neg. Reactivity → RFI	−0.015	0.008	[−0.033, −0.001]	−1.878	0.060
Uselessness Beliefs → Pos. Reactivity → RFI	−0.002	0.006	[−0.014, 0.010]	−0.255	0.798
Uselessness Beliefs → Neg. Reactivity → RFI	0.011	0.009	[−0.007, 0.029]	1.192	0.233
<i>Total Effects</i>					
Total Uncontrollability Beliefs Effect	−0.037	0.016	[−0.068, −0.007]	−2.359	0.018
Total Uselessness Beliefs Effect	−0.009	0.018	[−0.045, 0.026]	−0.520	0.603

Note. Emotion belief variables were assessed at baseline (T1). Emotional reactivity and resilient functioning variables were assessed at both T2 and T3. Reported coefficients are unstandardized estimates from the constrained model, in which corresponding structural paths were constrained to equality across follow-up assessments. RFI = Resilient Functioning Index.

2023). The primary analyses also provided some evidence consistent with previous findings that more maladaptive beliefs about the controllability of emotions are associated with greater negative emotional reactivity (Becerra et al., 2020). More robustly, the current findings showed that reduced positive and increased negative emotional reactivity were associated with poorer resilient functioning. However, the attenuation of the associations involving uncontrollability beliefs after accounting for baseline mental health suggests that their unique contribution to subsequent emotional reactivity and resilient functioning remains uncertain.

Nevertheless, the current study extends existing research on emotion beliefs and emotional reactivity by examining these processes in relation to individual differences in resilient functioning. Resilience is conceptually distinct from mental health, as it critically requires exposure to adversity and consideration of this exposure in its operationalisation (Chmitorz et al., 2018; Kalisch et al., 2021; Southwick et al., 2014). The current findings therefore offer a complementary perspective on the relationship between emotion-related processes and adaptive functioning following adversity, while highlighting the need for further research to establish whether emotion beliefs contribute to resilient

functioning independently of pre-existing mental health.

The possible contribution of maladaptive beliefs about emotions in variation in student resilience highlights a potential target for further investigation. The transition to university represents a critical period in the development of mental health difficulties, particularly for students experiencing financial stress, loneliness, and negative social interactions (Duffy et al., 2020; Farrer et al., 2026). Although the present findings do not establish that emotion beliefs provide incremental value for identifying students at risk beyond existing indicators of mental health, further research could examine whether particular emotion beliefs help identify vulnerability to poorer functioning following adversity. Establishing such incremental predictive value would be necessary before incorporating emotion belief assessments into university screening or prevention approaches. If supported, this could complement precision-based approaches involving early identification of risk and targeted interventions aimed at promoting resilience among students at heightened risk (e.g. Kuhlmeier et al., 2021).

Our findings provide some support for theoretical accounts of resilience which separately model resilience factors as stable individual characteristics that influence how stress is appraised, and resilience mechanisms as proximal processes that are engaged during or after stressor exposure (Kalisch et al., 2024). Emotion beliefs are thought to be a more stable individual difference dimension (Becerra et al., 2020) and therefore represent a candidate resilience factor influencing emotion regulation when exposed to adversity. In turn, emotional reactivity can vary not only person-to-person but also depending on stressor exposure (Howland et al., 2017) and thereby represents a plausible resilience mechanism. The observed associations between emotional reactivity and resilient functioning are consistent with this theoretical distinction, although the attenuation of associations involving emotion beliefs after accounting for baseline mental health means that their role as an independent resilience factor remains to be established. Moreover, these relationships are likely to be at least partly reciprocal over development, with repeated emotional experiences potentially contributing to the formation and reinforcement of emotion beliefs over time. Future research using cross-lagged longitudinal or experimental designs is needed to establish the causal and temporal pathways between these variables.

Should such causal pathways be established, it becomes relevant to consider how maladaptive beliefs about emotions could be therapeutically targeted. Cognitive Behavioural Therapy (CBT) has been shown to be effective in helping individuals challenge and change maladaptive belief patterns (Ryum & Nikolaos, 2024). Intervention programs have been developed that include techniques specifically targeting maladaptive beliefs about emotions (Corstorphine, 2006), and these methods have been shown to reduce symptoms of affective disorders (Farchione et al., 2024). However, more specific mechanistic evidence is needed to establish whether these techniques directly modify emotion-related beliefs and whether such changes subsequently promote resilient functioning.

The findings also highlight emotional reactivity as a potential intervention target, given its robust associations with resilient functioning. Interventions that reduce maladaptive negative emotional reactivity may therefore have potential for promoting resilient functioning, although the causal role of emotional reactivity in this relationship remains to be established. A randomized controlled longitudinal fMRI study found that socio-affective mental training involving loving-kindness meditation and dyadic exercises fostering compassion reduced emotional reactivity when processing images of human suffering, whereas attention-based training focused on present-moment interoceptive awareness and metacognitive training targeting perspective-taking and decentering did not (Favre et al., 2021). Future research could therefore investigate whether interventions targeting maladaptive emotion beliefs, potentially combined with approaches aimed at modifying emotional reactivity, can improve resilient functioning following adversity.

Given the established associations between emotion beliefs and mental health, and their potential role in individual differences in resilience, an important next step is to investigate how these beliefs develop and why some individuals hold more maladaptive emotion beliefs than others. One possible factor involves general beliefs about the self and the world. Research in adolescents has shown that general control beliefs (whether individuals believe they have control over important life circumstances) are positively associated with beliefs about the controllability of emotions (Ge et al., 2024). Other studies have shown that endorsing biomedical explanations for human behaviour such as attributing mental disorders to genetics can also influence one's beliefs and behaviours (Lebowitz & Appelbaum, 2019). Future research could therefore examine whether and how broader beliefs about the self and the world may drive maladaptive beliefs about emotions.

Of course, this study is not without its limitations. First, although the longitudinal design established temporal precedence of baseline emotion beliefs relative to subsequent emotional reactivity and resilient functioning, emotional reactivity and resilient functioning were assessed concurrently at each follow-up assessment. The study also remains correlational in nature and therefore does not allow conclusions regarding the causal relationships between these variables. Furthermore, associations involving uncontrollability beliefs were substantially attenuated after accounting for baseline mental health, indicating that the extent to which emotion beliefs independently contribute to subsequent emotional reactivity and resilient functioning remains uncertain. Future research using experimental designs and longitudinal assessments of emotion beliefs, emotional reactivity, and resilient functioning is needed to establish the direction and causal nature of these relationships.

Second, all constructs in the current study were assessed using self-report measures, which can be subject to limitations such as memory biases and low levels of introspective accuracy (Zimmerman, 2024). While such influences may be limited when assessing perceived mental health, emotional reactivity, and beliefs about emotions given these are inherently of subjective nature, future research could complement self-report data with alternative methodologies (e.g., psychophysiological measures for emotional reactivity) to examine converging evidence for these findings (Mauss et al., 2005).

Third, participants in the present study were not recruited based on clinical levels of psychological distress or elevated adversity exposure, and therefore it would be premature to assume that our findings generalize to clinical or highly trauma-exposed samples. However, our sample did demonstrate elevated levels of depressive and anxiety symptoms relative to normative young adult samples (Byrd-Bredbenner et al., 2021; Kliem et al., 2024), indicating that participants represented a relatively psychologically distressed student cohort. This may limit generalizability to less distressed student populations. At the same time, tertiary students experiencing elevated emotional distress are likely to be among those most in need of resilience-promoting interventions, making it clinically and practically important to understand the psychological processes associated with resilient functioning in such groups. Our sample was also predominantly female (representative of an undergraduate Psychology cohort) which should be taken into account in the interpretation of findings. Future studies could stratify by gender, adversity exposure, or clinical status to address these limitations, as well as examining the generalizability of our results to other samples.

Fourth, due to attrition across assessment points, the final sample size was relatively small. A Monte Carlo power analysis for the individual indirect effects (Schoemann et al., 2017) based on simplified parallel mediation models corresponding to each emotion-belief dimension, suggested that the power to detect the indirect path between uncontrollability beliefs and resilience via positive reactivity was

0.38, and via negative reactivity 0.56. Equivalent indirect effects for uselessness beliefs showed lower power (0.06 and 0.25, respectively). These findings suggest that some small indirect effects may have been underpowered, and replication with larger samples is warranted. Additionally, although the three-wave design allowed us to model resilient functioning relative to stressor exposure across a semester, this observation window remains relatively short. As a result, the RFI should be interpreted as indexing short to medium term resilient functioning rather than long term resilience. Future studies with longer follow up periods are needed to determine whether these patterns generalize to more sustained resilience over time.

Finally, the study was not preregistered. Although the hypotheses were theory-driven, preregistration would have strengthened transparency regarding analytic decisions and reduced researcher degrees of freedom.

In summary, this study highlights emotional reactivity as an important correlate of resilient functioning among tertiary students and provides preliminary evidence that beliefs about the controllability of emotions may also be relevant to individual differences in resilience. Although associations involving emotion beliefs were attenuated after accounting for baseline mental health, the findings identify important directions for further research into how relatively stable psychological characteristics and more proximal emotional processes may jointly contribute to resilient functioning. Establishing the causal and unique contribution of emotion beliefs will require further longitudinal and experimental research with larger samples. Such research may ultimately help identify modifiable psychological processes that can be targeted to support adaptive functioning among tertiary students experiencing adversity.

CRedit authorship contribution statement

Lies Notebaert: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Raffael Kalisch:** Writing – review & editing, Methodology, Conceptualization. **Mahdi Mazidi:** Writing – review & editing, Methodology, Investigation.

Ethics declaration

Written informed consent to take part in the study and to publish the article has been obtained from all participants or their legal representatives. The privacy rights of participants have been observed.

This study was performed in compliance with relevant laws, regulatory frameworks and guidelines where the research took place. This study was approved by the UWA Human Ethics Research Office. (Approval No. 2021/ET000074)

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the lead author used ChatGPT (OpenAI) to assist with coding, language editing, improving clarity and readability of the manuscript text, and formatting of written material. All scientific content, interpretations, and conclusions were determined by the authors. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interests

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. The authors have no conflict of interest.

Appendix A. Sensitivity Analyses Controlling for Baseline Mental Health

To examine whether the observed relationships remained evident after accounting for baseline mental health, an additional constrained single-group structural equation model (SEM) was estimated. As in the primary model, baseline emotion beliefs were specified as predictors of positive and negative emotional reactivity and resilient functioning at both T2 and T3. Baseline mental health was additionally included as a predictor of each of these outcomes, thereby estimating the associations between emotion beliefs, emotional reactivity, and resilient functioning while accounting for baseline mental health. Corresponding structural paths were constrained to equality across the two follow-up assessments, and cross-wave residual covariances between corresponding T2 and T3 measures were freely estimated to account for within-person dependencies across repeated assessments.

The model was estimated in lavaan (Rosseel, 2012) using full information maximum likelihood (FIML) to handle missing data. Standard errors and 95% confidence intervals were estimated using nonparametric bootstrap resampling with 10,000 bootstrap draws.

Baseline mental health significantly predicted lower positive emotional reactivity, greater negative emotional reactivity, and poorer resilient functioning at follow-up. As can be seen in Table A1, after controlling for baseline mental health, neither uncontrollability nor uselessness beliefs were significantly associated with emotional reactivity or resilient functioning, and none of the hypothesised indirect effects through emotional reactivity were statistically significant. The total effects of both uncontrollability and uselessness beliefs were also non-significant. Thus, the associations between emotion beliefs and subsequent outcomes observed in the primary model were substantially attenuated after accounting for baseline mental health.

Table A1

Unstandardized path coefficients, bootstrap standard errors (SE), 95% confidence intervals (CI), z-values, and p-values for the constrained structural equation model examining associations between baseline emotion beliefs and follow-up emotional reactivity and resilient functioning after controlling for baseline mental health symptoms.

Path	B	SE	95% CI	z	p
<i>Direct Effects on Resilient Functioning Index</i>					
Uncontrollability Beliefs	−0.008	0.009	[−0.027, 0.009]	−0.855	0.393
Uselessness Beliefs	−0.019	0.011	[−0.041, 0.003]	−1.689	0.091
Positive Reactivity	0.025	0.004	[0.017, 0.033]	5.97	< 0.001
Negative Reactivity	−0.032	0.004	[−0.039, −0.023]	−7.385	< 0.001
Baseline Mental Health	−0.066	0.022	[−0.109, −0.022]	−2.973	0.003
<i>Paths Predicting Positive Reactivity</i>					
Uncontrollability Beliefs	−0.105	0.184	[−0.451, 0.281]	−0.568	0.57
Uselessness Beliefs	−0.145	0.194	[−0.538, 0.220]	−0.748	0.455
Baseline Mental Health	−1.332	0.348	[−1.985, −0.608]	−3.831	< 0.001
<i>Paths Predicting Negative Reactivity</i>					
Uncontrollability Beliefs	0.022	0.196	[−0.313, 0.457]	0.11	0.912
Uselessness Beliefs	−0.183	0.192	[−0.593, 0.151]	−0.955	0.339
Baseline Mental Health	2.865	0.321	[2.198, 3.459]	8.929	< 0.001
<i>Indirect Effects on Resilience</i>					
Uncontrollability Beliefs → Pos. Reactivity → RFI	−0.003	0.005	[−0.012, 0.007]	−0.561	0.575
Uncontrollability Beliefs → Neg. Reactivity → RFI	−0.001	0.006	[−0.014, 0.010]	−0.111	0.912
Uselessness Beliefs → Pos. Reactivity → RFI	−0.004	0.005	[−0.014, 0.005]	−0.746	0.455
Uselessness Beliefs → Neg. Reactivity → RFI	0.006	0.006	[−0.005, 0.019]	0.944	0.345
<i>Total Effects</i>					
Total Uncontrollability Beliefs Effect	−0.011	0.012	[−0.036, 0.011]	−0.917	0.359
Total Uselessness Beliefs Effect	−0.016	0.014	[−0.043, 0.011]	−1.199	0.23

Note. Emotion belief variables were assessed at baseline (T1). Emotional reactivity and resilient functioning variables were assessed at both T2 and T3. Reported coefficients are unstandardized estimates from the constrained model, in which corresponding structural paths were constrained to equality across follow-up assessments. RFI = Resilient Functioning Index.

Data availability

The data is available on OSF (link included in manuscript)

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